

Paramjeet Rout

Generative AI Engineer & Data Analyst | RAG pipelines, multi-agent LLM systems, reliable analytics

Gurugram / India | routparamjeet@gmail.com | github.com/routparam12 | linkedin.com/in/paramjeet-rout2026

Summary

Generative AI Engineer and Data Analyst with 3+ years across data analytics and applied AI. Builds production RAG pipelines, multi-agent orchestration, LLM integrations, ETL workflows, and dashboards that make messy data dependable and useful.

Experience

Generative AI Engineer / Data Analyst - Unimrkt Response LLP, Gurugram (Hybrid)

Oct 2025 - Present

- Architecting a production RAG chatbot upgrade using LangChain and LangGraph to retrieve context from relational databases and JSON sources via FAISS.
- Delivered a fraud detection system with fuzzy matching and Decision Tree classification, achieving 70% precision and tracking experiments in MLflow.
- Designed PostgreSQL to Elasticsearch ETL via Logstash with 3-minute incremental refresh for warehouse-scale analysis without production load.
- Iterated on token optimization, chunk sizing, and retrieval thresholds to balance LLM accuracy and API cost in production.

Data Scientist Associate - Tech Mahindra, Ahmedabad

Mar 2024 - May 2025

- Built PDF parsing, metadata extraction, and data cleaning automation with Python and Azure Functions.
- Integrated OpenAI, Gemini, and Together AI behind a unified abstraction layer for fallback and cost-optimized inference.
- Automated XML test report processing with Python and PowerShell, reducing compile time from 4 hours to 1 hour.
- Reduced manual reporting effort by 30% through optimized DAX measures and data models.

Selected Projects

Dora.ChatBot (SurveySenseAI) - FastAPI, LangChain, LangGraph, Hugging Face, FAISS, Redis, PostgreSQL, Docker, Azure Pipelines

Full-stack Generative AI customer support platform with multi-agent orchestration, vector retrieval, model fallback, persistent memory, JWT auth, and multi-source ingestion.

PaperShield - FastAPI, LangGraph, LlamaIndex, ChromaDB, Gemini API

Leak-resistant exam engine that generates MCQs dynamically from syllabus PDFs at exam time, avoiding stored fixed question banks.

Fraud User Detection System - Python, Scikit-learn, Fuzzy Logic, MLflow, Streamlit

Duplicate and high-risk account detection using phonetic and edit-distance scoring, browser fingerprinting, behavioral signals, and MLflow-tracked evaluation.

Skills

Generative AI	LangChain, LangGraph, LlamaIndex, RAG, multi-agent systems, OpenAI GPT, Gemini API
Data & Search	Python, SQL, PostgreSQL, Elasticsearch, Logstash, ETL, FAISS, ChromaDB, Pinecone
ML & Analytics	Scikit-learn, Decision Tree, fuzzy matching, feature engineering, MLflow, Power BI, DAX
Backend	FastAPI, JWT authentication, SQLAlchemy ORM, Alembic, Streamlit, Docker, Azure Pipelines